

EXHIBIT 19

Ex. 19 - Infringement Contention Claim Chart
U.S. Patent No. 12,456,334 (“the ’334 patent”)

Apple’s products directly and/or indirectly infringe at least claims 1, 7, 11, 13, 15 (“Asserted Claims”) of U.S. Patent No. 12,456,334 (“the ’334 patent”). The products that infringe the ’334 patent include at least the iPhone 15, iPhone 15 Plus, iPhone 15 Pro, iPhone 15 Pro Max, iPhone 16, iPhone 16e, iPhone 16 Pro, iPhone 16 Pro Max, iPhone 16 Plus, iPhone 17, iPhone 17e, iPhone 17 Pro, iPhone 17 Pro Max, iPhone Air, 11-inch iPad Pro (4th Generation), iPad Pro 12.9 inch (6th Generation), iPad Pro 11-inch (M4 and M5), iPad Pro 13-inch (M4 and M5) and other infringing iPhone and iPad models (each an “Accused Apple Product” and collectively the “Accused Apple Products”). The Accused Apple Products infringe the ’334 patent as set forth below. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

Additionally, the Apple iPhone 17 is representative of the Accused Apple Products at least because the other Accused Apple Products function in the same or substantially similar manner to the Apple iPhone 17.¹ The relevant identification processes and functionalities, as well as the hardware components of the Apple iPhone 17, are demonstrated through testing of the Apple iPhone 17 (“Apple Product Testing”), which is available for inspection upon request.

Claim 1	Accused Apple Products
[1pre] A system, comprising:	The Accused Apple Products comprise a system. For example, the Apple iPhone 17 is a system:

¹ Each of the Accused Apple Products includes the claimed hardware components and steps. The hardware components and source code implementing the relevant functionalities of the Apple iPhone 17 are representative of the Accused Apple Products for the purpose of infringement. To the extent there may be differences in the hardware composition, certain software steps, or source code versions among these Accused Apple Products, those differences are not material for the infringement analysis.

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	 Exhibit 36.
[1a] a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising an optical sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength;	The Accused Apple Products comprise a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising an optical sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image.” Exhibit 35, at 22. The Apple iPhone 17 Face ID module includes a near-infrared camera module.

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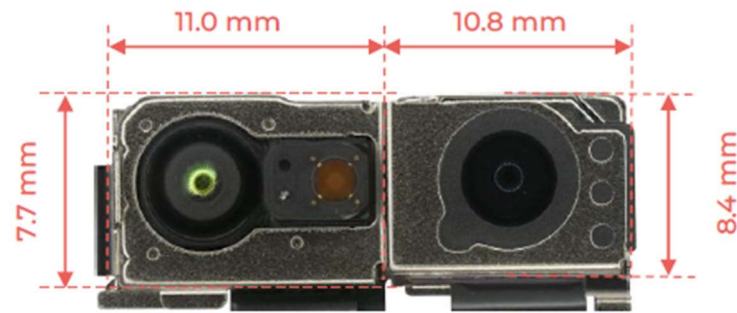
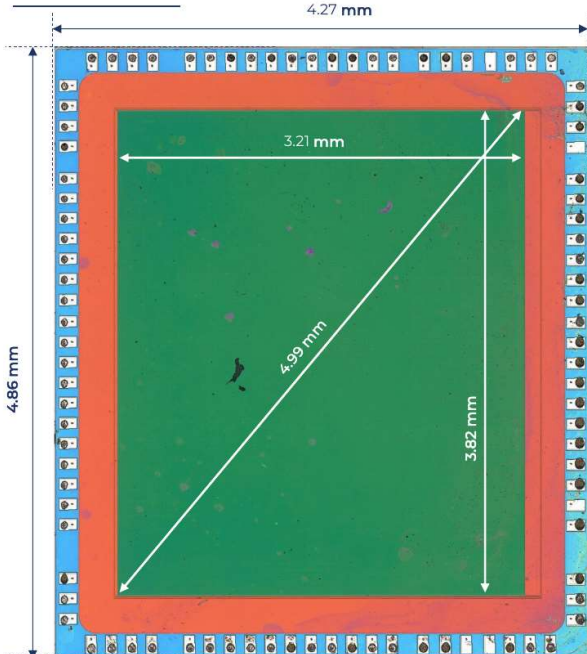
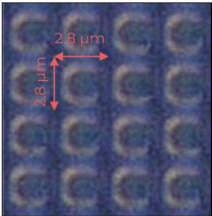


Exhibit 34C, at 26.

The near-infrared camera uses a monochrome NIR CMOS image sensor.

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	NIR CMOS IMAGE SENSOR DIE Overview  - Die area: 20.75 mm² (4.86 x 4.27 mm) - Nb of PGDW per 12-inch wafer: 3,076 - Pad number: 94 - Pixel array area: 12.26 mm² (3.82 x 3.21 mm) - Pixel size: 7.84 μm² (2.8 x 2.8 μm) - Resolution: 1.56 Mp (1,364 x 1,146) - Color Filter: Monochrome - Pattern: - Marking: S6G812A1A   ipugan@trinamix.de Die Overview - Optical View ©2026 Yole Group SPR26034 Apple iPhone 17 Face ID www.yolegroup.com 46
[1b] a plurality of light projecting devices at least one of which is configured to project a light pattern on the scene, wherein at least one of the plurality of light projecting devices is a vertical cavity surface emitting laser (VCSEL); and	The Accused Apple Products comprise a plurality of light projecting devices at least one of which is configured to project a light pattern on the scene, wherein at least one of the plurality of light projecting devices is a vertical cavity surface emitting laser (VCSEL). The TrueDepth camera system includes a dot projector Vertical-Cavity Surface-Emitting Laser (“VCSEL”), a flood illumination VCSEL, and a near-infrared camera module.

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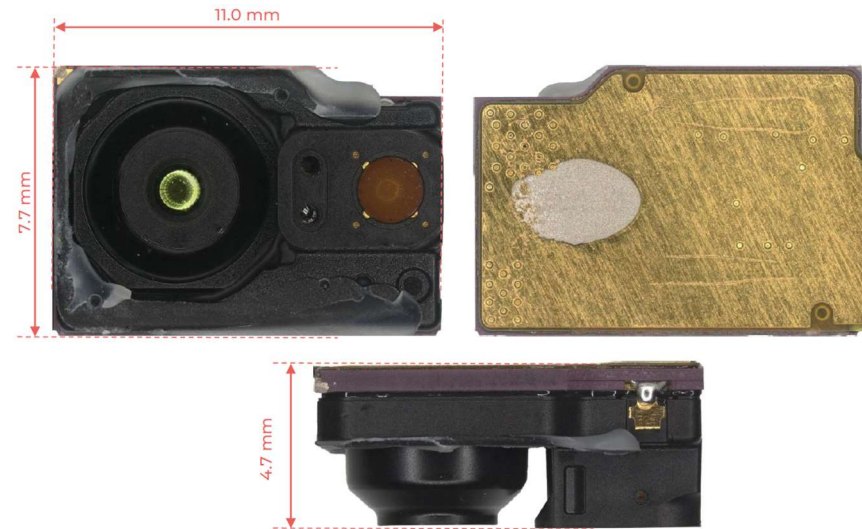


Exhibit 34C, at 35.

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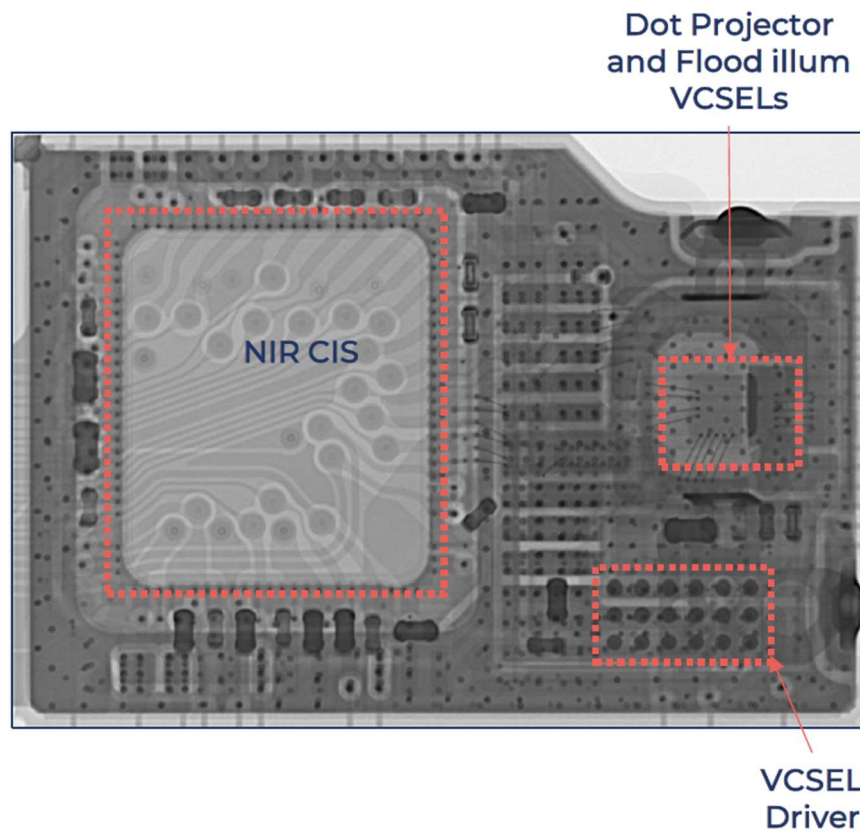
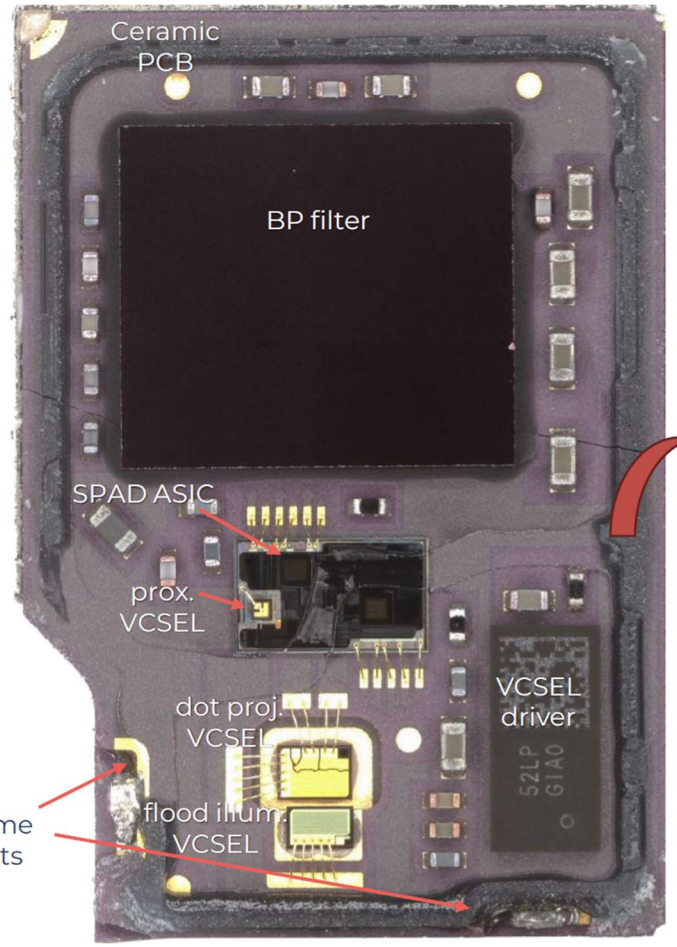


Exhibit 34C, at 36.

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	iPhone 17 Pro Face ID Module Opened © Yole Group 2026  The photograph shows the internal components of an iPhone 17 Pro Face ID module. The components are labeled as follows: Ceramic PCB (top), BP filter (large black rectangle), SPAD ASIC (small chip), prox. VCSEL (small chip), dot proj. VCSEL (small chip), flood illum. VCSEL (small chip), DOE leadframe contacts (bottom left), VCSEL driver (chip labeled 52LP 61A0), and a red curved arrow pointing to the right side of the module.
	Exhibit 34C, at 37.
[1c] a processor configured to:	The Accused Apple Products comprise a processor. For example, the Apple iPhone 17 includes a processing chip.

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	Chip  A19 chip 6-core CPU with 2 performance and 4 efficiency cores 5-core GPU with Neural Accelerators 16-core Neural Engine Hardware-accelerated ray tracing Exhibit 33.
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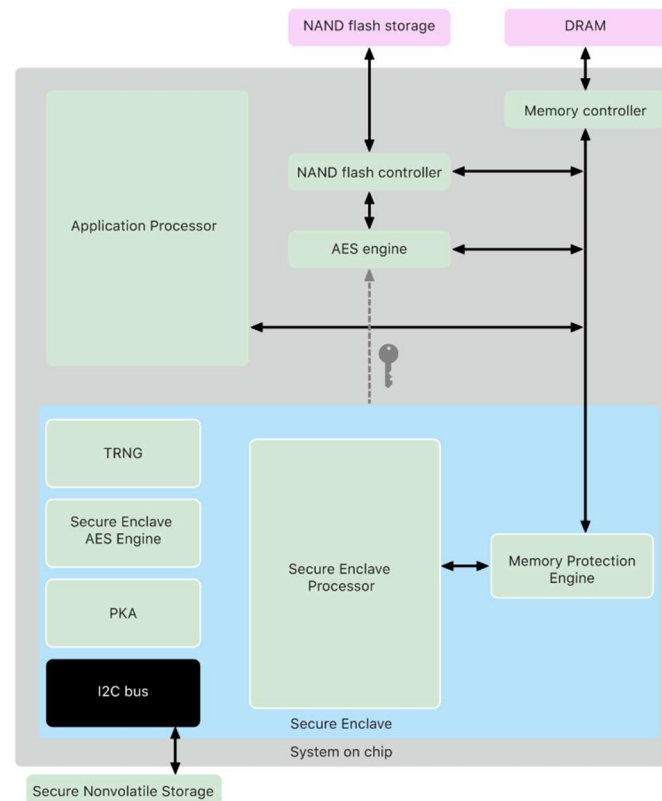
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Secure Enclave

The Secure Enclave is a dedicated secure subsystem in the latest versions of iPhone, iPad, Mac, Apple TV, Apple Watch, Apple Vision Pro, and HomePod.

Overview

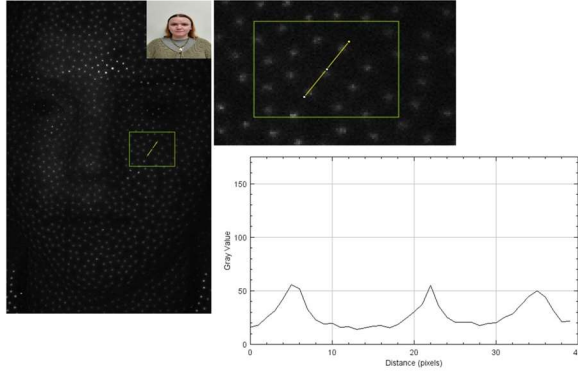
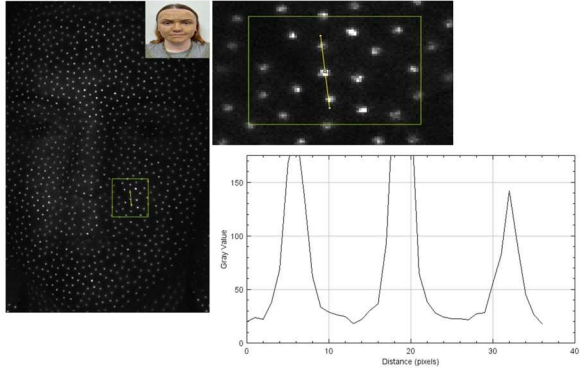
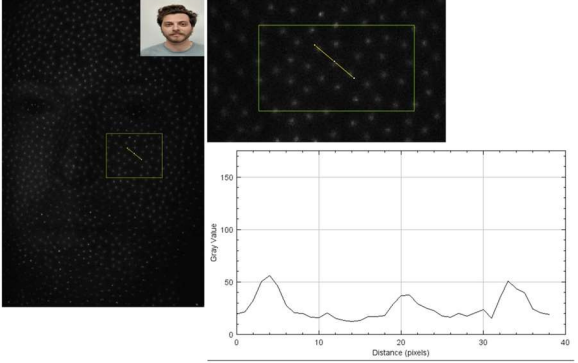
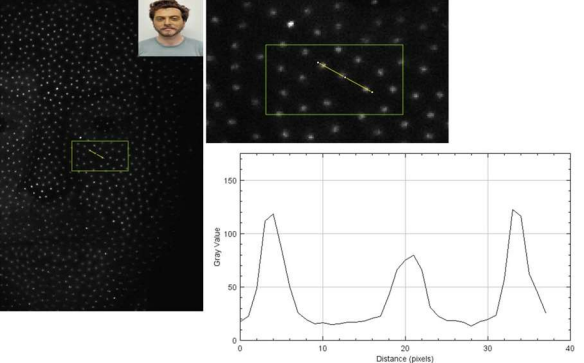
The Secure Enclave is a dedicated secure subsystem integrated into Apple [system on a chip \(SoC\)](#). The Secure Enclave is isolated from the main processor to provide an extra layer of security and is designed to keep sensitive user data secure even when the Application Processor kernel becomes compromised. It follows the same design principles as the SoC does—a boot ROM to establish a hardware root of trust, an AES engine for efficient and secure cryptographic operations, and protected memory. Although the Secure Enclave doesn’t include storage, it has a mechanism to store information securely on attached storage separate from the NAND flash storage that’s used by the Application Processor and operating system.



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	Exhibit 35, at 10.
[1d] output a first image comprising light reflected from the scene in response to the projection by the light projecting devices and output a second image comprising a reflective light pattern reflected from the scene in response to the projection by the light projecting devices;	The Accused Apple Products comprise a processor configured to output a first image comprising light reflected from the scene in response to the projection by the light projecting devices and output a second image comprising a reflective light pattern reflected from the scene in response to the projection by the light projecting devices. “The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. For example, Face ID on the Apple iPhone 17 outputs multiple images comprising light reflected from the scene in order to detect a face in the images.

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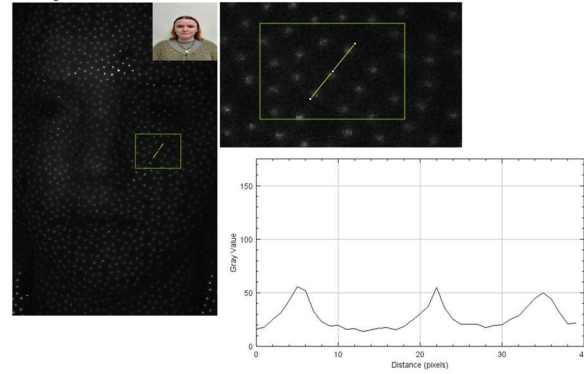
	Subject01 Covered area: none – natural Face  Subject01 Covered area: entire face  Subject02 Covered area: none – natural Face  Subject02 Covered area: entire face  Exhibit 47 ¶¶ 23-31.
[1e] detect a face in a region in the first image;	The Accused Apple Products comprise a processor configured to detect a face in a region in the first image. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses

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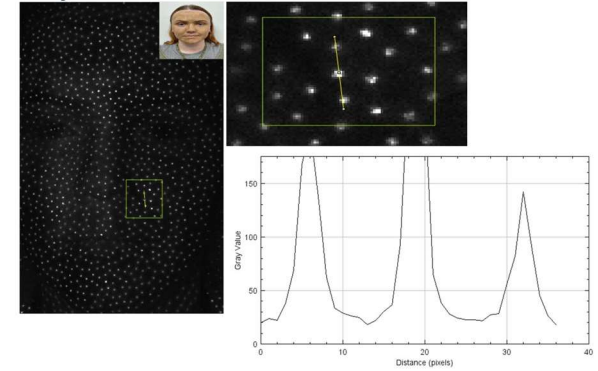
	advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.” Exhibit 35, at 22. “The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device” Exhibit 35, at 22.
[1f] determine a distribution of light intensity of a region in the second image;	The Accused Apple Products comprise a processor configured to determine a distribution of light intensity of a region in the second image. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. For example, Face ID on the Apple iPhone 17 succeeds when given certain distributions of light intensity of a first subregion inclusive of the geometric center of the region, where the region constitutes the area around a dot, but fails when given other distributions. The different distributions constitute different shapes and brightnesses.

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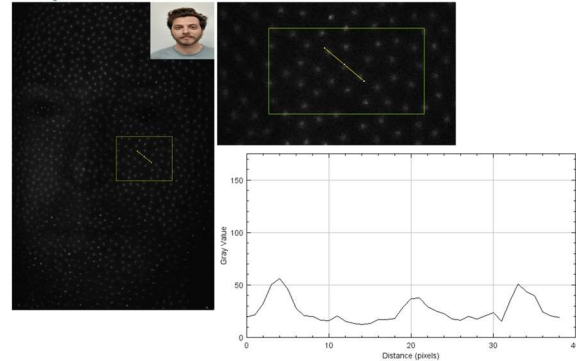
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

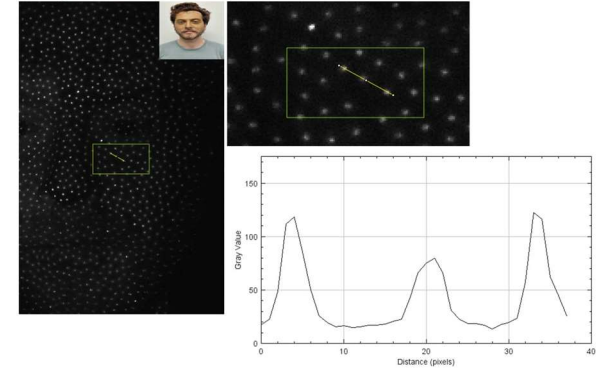


Exhibit 47 ¶¶ 23-31.

Additionally and alternatively, Face ID works “even with a mask on,” indicating that Face ID determines distributions of particular facial regions.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

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	Exhibit 35, at 22.
[1g] compare the determined distribution of light intensity to a distribution of light intensity derived from a reference image;	The Accused Apple Products comprise a processor configured to compare the determined distribution of light intensity to a distribution of light intensity derived from a reference image. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. “Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

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Biometric security

Passcodes and passwords are essential to the security of Apple devices. At the same time, users require convenient access to their devices, often more than a hundred times a day. Biometric authentication provides a way to retain the security of a strong passcode—or even strengthen the passcode or password because it need not be entered manually—while providing the convenience of swiftly unlocking with a finger press or glance. Optic ID, Face ID, and Touch ID don’t replace a passcode or password, but in most situations they do make access faster and easier.

Apple’s biometric security architecture relies on a strict separation of responsibilities between the biometric sensor and the Secure Enclave, and a secure connection between the two. The sensor captures the biometric image and securely transmits it to the Secure Enclave. During enrollment, the Secure Enclave processes, encrypts, and stores the corresponding Optic ID, Face ID, and Touch ID template data. During matching, the Secure Enclave compares incoming data from the biometric sensor against the stored templates to determine whether to unlock the device or respond that a match is valid (for Apple Pay, in-app, and other uses of Optic ID, Face ID, and Touch ID). The architecture supports devices that include both the sensor and Secure Enclave (such as iPhone, iPad, many Mac computers, and Apple Vision Pro), as well as the ability to physically separate the sensor into a peripheral that’s then securely paired to the Secure Enclave in a Mac with Apple silicon.

Exhibit 35, at 20.

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Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

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Exhibit 35, at 22.

Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

Exhibit 35, at 29.

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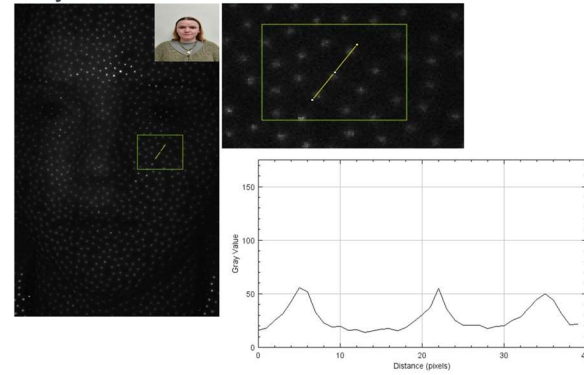
	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device’s neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It’s designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[1h] determine a material classification of the face based on the comparison of the determined distribution of light intensity and the distribution of light intensity derived from the reference image;	The Accused Apple Products comprise a processor configured to determine a material classification of the face based on the comparison of the determined distribution of light intensity and the distribution of light intensity derived from the reference image. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22.

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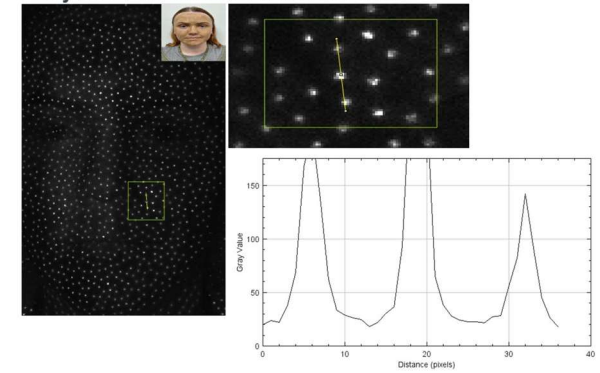
	“After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. “Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29. For example, Face ID on the Apple iPhone 17 succeeds when imaging faces of certain materials but fails when imaging faces of other materials, based on reference data.
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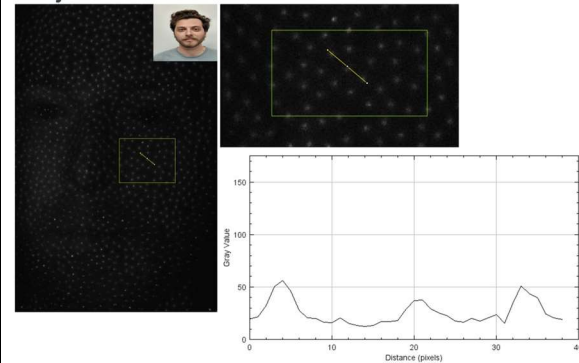
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

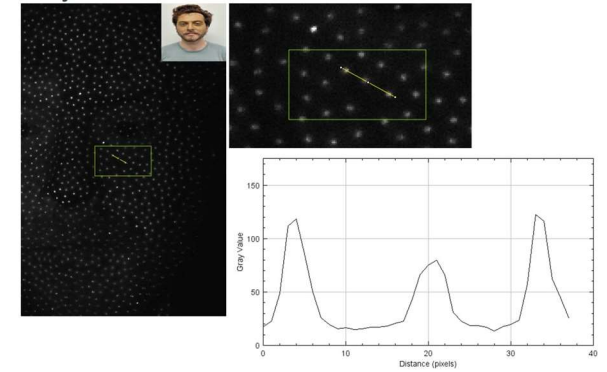


Exhibit 47 ¶¶ 23-31.

Face ID further uses neural networks to compare the given regions to corresponding regions in reference image data.

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Face ID security

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Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

Ex. 19 - Infringement Contention Claim Chart
U.S. Patent No. 12,456,334 (“the ’334 patent”)

Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

Exhibit 35, at 29.

Ex. 19 - Infringement Contention Claim Chart
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	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It's designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[1i] determine at least one depth map based on depth differences in the reflective light pattern; and	The Accused Apple Products comprise a processor configured to determine at least one depth map based on depth differences in the reflective light pattern. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and

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	reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.” Exhibit 35, at 22. “Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
[1j] authenticate the face based on the detection of the face, the classification of the face, and the depth map.	The Accused Apple Products comprise a processor configured to authenticate the face based on the detection of the face, the classification of the face, and the depth map. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map

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	captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.” Exhibit 35, at 22. “Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
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